Syllabus Details

Syllabus ID:	14177
Syllabus Name:	**Software development project_Dự án phát triển phần mềm**
Course Name English:	**Software Development Project**
Subject Code:	**SWP391**
Learning-Teaching Method:	Lecture, Lab, Group Discussion, Presentation, Peer Review.AI-integrated tasks across all Software Development Life Cycle phases
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h Project Presentation + 104h self-study
Pre-Requisite:	PRJ301, SWE201c or SWE202c, pass LAB211
Description:	This course guides students through the full Software Development Life Cycle (SDLC) by working on a real-world team project, with an emphasis on applying AI responsibly in requirement analysis, design, implementation, and testing. Students will practice user story writing, system design, coding using MVC and OOP, workflow development, testing, and reporting. AI tools are integrated into each phase to enhance productivity and quality.
StudentTasks:	- Join the team as arranged by the teacher - Select the project topic & technology stack and get approval from teacher before implementing - Attend at least 80% of contact slots in order to be accepted to the final presentation. - Analyze requirements, write user requirements, and create system diagrams with AI-assisted tools. - Design system architecture, database schema, and UI/UX wireframes using MVC, OOP, and AI suggestions. - Implement core features with AI-supported coding, apply coding standards, and manage source code in Git. - Develop and run unit/integration tests, using AI to suggest and refine test cases. - Deploy the application to staging, verify functionality, and prepare for final delivery. - Manage project progress, present milestones, and prepare high-quality reports with AI support.
Tools:	- Requirement & Design: Miro, Lucidchart, Figma, Draw.io, ChatGPT, Gemini - Coding & Debugging: IntelliJ IDEA / VS Code / Visual Studio, GitHub Copilot, Tabnine - Database: MySQL Workbench, SQL Server Management Studio, pgAdmin -Testing: JUnit, NUnit, Postman, Selenium, AI-powered test case generators - Project Management: Jira, Trello, Notion, Git/GitHub - Deployment: Docker, Heroku, AWS / Azure / GCP, Vercel
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	377/QĐ-ĐHFPT dated 04/09/2026
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2026

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Prompt Engineering for Developers (OpenAI & DeepLearning.AI)	Deep Learning AI	Coursera	N/A	N/A		True	False	True	https://www.deeplearning.ai/short-courses/prompt-engineering/
2	Software Engineering	Ian Sommerville	Pearson	2015	10th	978-0133943030	False	True	False	https://www.amazon.com/dp/013394303

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Spring MVC With Spring Boot (All Modules) and Project	Packt - Course Instructors	Coursera	N/A	N/A		False	False	True	https://www.coursera.org/learn/packt-spring-mvc-with-spring-boot-all-modules-and-project-xlnjc
4	ASP.NET Core MVC Fundamentals & CRUD Operations	Packt - Course Instructors	Coursera	N/A	N/A		False	False	True	https://www.coursera.org/learn/packt-asp-net-core-mvc-fundamentals-crud-asd1a?specialization=packt-asp-net-core-mvc-net-8-the-complete-guide

6 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Analyze and clarify project requirements with AI-assisted elicitation and documentation
2	CLO2	Design system modules and databases using MVC, OOP, and AI-supported modelling tools
3	CLO3	Create ERD models and implement relational databases with SQL, applying AI tools for design validation and query optimization.
4	CLO4	Develop web applications in Java/.NET/Node.js with AI-enhanced coding, debugging, and workflow implementation.
5	CLO5	Demonstrate teamwork, professional ethics, and responsible AI use in software development.
6	CLO6	Deliver effective presentations and reports with AI-aided content and visualization.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction & Topic Allocation	Lecture, Discussion	CLO1, CLO4	AI literacy, Collaboration	Course outline, IDE	SWP391	Listen to course overview, join teams	
2	Course Introduction & Topic Allocation	Lecture, Discussion	CLO1, CLO4	AI literacy, Collaboration	Course outline, IDE	SWP391	Listen to course overview, join teams	
3	Introduction to AI-Augmented SDLC	Lecture, Discussion	CLO1, CLO4	AI literacy, Collaboration	Reading file, IDE		Learn SDLC concept, note AI examples	
4	User Requirements Development with Domain Experts (invited guest, lecturer role-play or AI Agents).	Lecture, Lab,Discussion	CLO1, CLO4	AI literacy, Problem solving	Requirement doc, IDE		Practice requirement elicitation, draft simple requirement document	
5	User Requirements Development with Domain Experts (invited guest, lecturer role-play or AI Agents).	Lecture, Lab,Discussion	CLO1, CLO4	AI literacy, Problem solving	Requirement doc, IDE		Practice requirement elicitation, draft simple requirement document	
6	User Requirements Development with Domain Experts (invited guest, lecturer role-play or AI Agents).	Lecture, Lab,Discussion	CLO1, CLO4	AI literacy, Problem solving	Requirement doc, IDE		Practice requirement elicitation, draft simple requirement document	
7	Data Modeling- ERD Conceptual & Logical	Lecture, Lab,Discussion	CLO1, CLO4	AI literacy, Problem solving	ERD tool, diagrams		Sketch ER diagram, refine with relationships, validate with data	
8	Data Modeling- ERD Conceptual & Logical	Lecture, Lab,Discussion	CLO1, CLO4	AI literacy, Problem solving	ERD tool, diagrams		Sketch ER diagram, refine with relationships, validate with data	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
9	Data Modeling- ERD Conceptual & Logical	Lecture, Lab,Discussion	CLO1, CLO4	AI literacy, Problem solving	ERD tool, diagrams		Sketch ER diagram, refine with relationships, validate with data	
10	Database design- Tables and Constraints	Lecture, Discussion	CLO1, CLO2	Critical thinking, AI literacy	SQL scripts, IDE		Map ERD to tables, add keys/constraints, run sample DDL	
11	Database design- Tables and Constraints	Lecture, Discussion	CLO1, CLO2	Critical thinking, AI literacy	SQL scripts, IDE		Map ERD to tables, add keys/constraints, run sample DDL	
12	Database design- Tables and Constraints	Lecture, Discussion	CLO1, CLO2	Critical thinking, AI literacy	SQL scripts, IDE		Map ERD to tables, add keys/constraints, run sample DDL	
13	Software design with MVC Pattern	Lecture, Lab	CLO2	Creativity, AI literacy	MVC design doc, IDE		Sketch MVC design, define controllers/views, present diagram	
14	Software design with MVC Pattern	Lecture, Lab	CLO2	Creativity, AI literacy	MVC design doc, IDE		Sketch MVC design, define controllers/views, present diagram	
15	Software design with MVC Pattern	Lecture, Lab	CLO2	Creativity, AI literacy	MVC design doc, IDE		Sketch MVC design, define controllers/views, present diagram	
16	UI & Workflow Prototyping	Lecture, Lab	CLO2	Creativity, AI literacy	Prototyping tool, UI mockups		Draw wireframes, build prototype, present to peers	
17	UI & Workflow Prototyping	Lecture, Lab	CLO2	Creativity, AI literacy	Prototyping tool, UI mockups		Draw wireframes, build prototype, present to peers	
18	UI & Workflow Prototyping	Lecture, Lab	CLO2	Creativity, AI literacy	Prototyping tool, UI mockups		Draw wireframes, build prototype, present to peers	
19	Milestone 1 Review & Evaluation	Presentation, Q&A	CLO1, CLO2	Communication, Critical thinking	Presentation slides, report		Present milestone, receive feedback	
20	Milestone 1 Review & Evaluation	Presentation, Q&A	CLO1, CLO2	Communication, Critical thinking	Presentation slides, report		Present milestone, receive feedback	
21	Milestone 1 Review & Evaluation	Presentation, Q&A	CLO1, CLO2	Communication, Critical thinking	Presentation slides, report		Present milestone, receive feedback	
22	Development Phase Planning (Phase 2)	Lecture, Lab	CLO2, CLO2	AI literacy, Problem solving	Planning doc, chart		Make phase plan, assign tasks, confirm timeline	
23	Environment Setup and Version Control	Lecture, Lab	CLO2, CLO3	AI literacy, Problem solving	Git client, IDE		Install IDE & Git, configure environment, practice commit/push	
24	Environment Setup and Version Control	Lecture, Lab	CLO2, CLO3	AI literacy, Problem solving	Git client, IDE		Install IDE & Git, configure environment, practice commit/push	
25	Build Model in MVC- Develop Repository & Service Layers	Lecture, Lab	CLO6	Problem solving, AI literacy	Git client, IDE		Write sample repository/service code, commit and test	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
26	Build Model in MVC- Develop Repository & Service Layers	Lecture, Lab	CLO5	Problem solving, AI literacy	Git client, IDE		Write sample repository/service code, commit and test	
27	Build Model in MVC- Develop Repository & Service Layers	Lecture, Lab	CLO4	Problem solving, AI literacy	Git client, IDE		Write sample repository/service code, commit and test	
28	Workflow 0: Basic Web Security- Register/Login/Logout/Forgot Password	Lecture, Lab	CLO0	Problem solving, AI literacy	Git client, IDE		Implement login form, add validation, connect DB	
29	Workflow 0: Basic Web Security- Register/Login/Logout/Forgot Password	Lecture, Lab	CLO1	Problem solving, AI literacy	Git client, IDE		Implement login form, add validation, connect DB	
30	Workflow 0: Basic Web Security- Register/Login/Logout/Forgot Password	Lecture, Lab	CLO2	Problem solving, AI literacy	Git client, IDE		Implement login form, add validation, connect DB	
31	Workflow 1: Basic CRUD in DB Tables- Master Data Setup	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Create CRUD forms, insert/update/test records	
32	Workflow 1: Basic CRUD in DB Tables- Master Data Setup	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Create CRUD forms, insert/update/test records	
33	Workflow 1: Basic CRUD in DB Tables- Master Data Setup	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Create CRUD forms, insert/update/test records	
34	Workflow 2: Core Transaction Flow (Success Path)	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Draw transaction flow, code success path, run tests	
35	Workflow 2: Core Transaction Flow (Success Path)	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Draw transaction flow, code success path, run tests	
36	Workflow 2: Core Transaction Flow (Success Path)	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Draw transaction flow, code success path, run tests	
37	Workflow 2: Core Transaction Flow (Exception Path)	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Draw transaction flow, code exception path, run tests	
38	Workflow 2: Core Transaction Flow (Exception Path)	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Draw transaction flow, code exception path, run tests	
39	Workflow 2: Core Transaction Flow (Exception Path)	Lecture, Lab	CLO3	Problem solving, AI literacy	Git client, IDE		Draw transaction flow, code exception path, run tests	
40	AI-support for Automation Testing	Presentation, Q&A	CLO2, CLO3	Communication, Critical thinking	Test cases, AI tool		Try AI tool for test cases, run scripts, adjust test scripts	
41	AI-support for Automation Testing	Presentation, Q&A	CLO2, CLO3	Communication, Critical thinking	Test cases, AI tool		Try AI tool for test cases, run scripts, adjust test scripts	
42	AI-support for Automation Testing	Presentation, Q&A	CLO2, CLO3	Communication, Critical thinking	Test cases, AI tool		Try AI tool for test cases, run scripts, adjust test scripts	
43	Milestone 2 Review & Evaluation	Presentation, Q&A	CLO2, CLO3	Communication, Critical thinking	Presentation slides, report		Present milestone, receive feedback	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
44	Milestone 2 Review & Evaluation	Presentation, Q&A	CLO2, CLO3	Communication, Critical thinking	Presentation slides, report		Present milestone, receive feedback	
45	Milestone 2 Review & Evaluation	Presentation, Q&A	CLO2, CLO3	Communication, Critical thinking	Presentation slides, report		Present milestone, receive feedback	
46	Workflow 3: Dashboard & Reporting	Lecture, Lab	CLO3	Problem solving, AI literacy	Dashboard mockup, KPI sheet		Identify KPIs, build charts, present dashboard	
47	Workflow 3: Dashboard & Reporting	Lecture, Lab	CLO3	Problem solving, AI literacy	Dashboard mockup, KPI sheet		Identify KPIs, build charts, present dashboard	
48	Workflow 3: Dashboard & Reporting	Lecture, Lab	CLO3	Problem solving, AI literacy	Dashboard mockup, KPI sheet		Identify KPIs, build charts, present dashboard	
49	Software Deployment and Data Preparation	Lecture, Discussion	CLO4, CLO5	AI literacy, Communication	Deployment script		Practice deployment steps, seed sample data, run demo	
50	Software Deployment and Data Preparation	Lecture, Discussion	CLO4, CLO5	AI literacy, Communication	Deployment script		Practice deployment steps, seed sample data, run demo	
51	Software Deployment and Data Preparation	Lecture, Discussion	CLO4, CLO5	AI literacy, Communication	Deployment script		Practice deployment steps, seed sample data, run demo	
52	User Acceptance Tesing and Quality Improvement	Presentation, Q&A	CLO5	Communication, Collaboration	UAT report, defect log		Join UAT, fix small defects, re-test improvements	
53	User Acceptance Tesing and Quality Improvement	Presentation, Q&A	CLO5	Communication, Collaboration	UAT report, defect log		Join UAT, fix small defects, re-test improvements	
54	User Acceptance Tesing and Quality Improvement	Presentation, Q&A	CLO5	Communication, Collaboration	UAT report, defect log		Join UAT, fix small defects, re-test improvements	
55	Milestone 3- Final Presentation	Presentation, Q&A	CLO5	Communication, Collaboration	Presentation slides, report		Present milestone, receive feedback	
56	Milestone 3- Final Presentation	Presentation, Q&A	CLO5	Communication, Collaboration	Presentation slides, report		Present milestone, receive feedback	
57	Milestone 3- Final Presentation	Presentation, Q&A	CLO5	Communication, Collaboration	Presentation slides, report		Present milestone, receive feedback	
58	Project Closing and Retrospective	Presentation, Q&A	CLO3, CLO4, CLO5	Communication, Critical thinking	Retrospective notes, repo archive		Join retrospective, clean up repo, share lessons learned	
59	Project Closing and Retrospective	Presentation, Q&A	CLO3, CLO4, CLO5	Communication, Critical thinking	Retrospective notes, repo archive		Join retrospective, clean up repo, share lessons learned	
60	Project Closing and Retrospective	Presentation, Q&A	CLO3, CLO4, CLO5	Communication, Critical thinking	Retrospective notes, repo archive		Join retrospective, clean up repo, share lessons learned	

0 Constructive question(s)

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assessment 1 (Week3)Milestone 1: Requirement Analysis & Design	on-going	1	15.0%	>0	20'/group	CLO1, CLO2, CLO4	The student presents slides, gives a demonstration, and answers questions from the lecturer.		Requirement analysing, AI-assisted documentation, system architecture, initial UI, ERD	Evaluation Items: - SRS Document (30%): ≥15 pages, ≥12 user stories, complete and validated with AI + expert. - Design Artifacts (50%): ERD with ≥8 entities, valid state diagram, clear architecture, consistent UI. - Process (20%): ≥6 AI logs and 4 weekly reports with evidence.	View Rubrics Docume for more details
2	Assessment 2 (Week 8)Milestone 2 : Workflow 1 & 2 Implementation	on-going	1	20.0%	>0	20'/group	CLO2, CLO3, CLO4	The student presents slides, gives a demonstration, and answers questions from the lecturer.		Coding modules, database integration, AI-assisted debugging, functional testing	Evaluation Items: - Workflow Implementation (50%): Workflow 0 + Workflow 1 + Workflow 2 (main flow + ≥2 exception paths) fully functional. - Codebase & Database (30%) : Multi-layer design (e.g., MVC), complete CRUD classes, DB ≥10 tables, normalized with full PK/FK. - Process (20%) : ≥8 commits/member + ≥6 AI debugging logs.	View Rubrics Docume for more details

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
3	Assessment 3Milestone 3 (Week 10): Full System Completion & Testing	on-going	1	25.0%	>0	20'/group	CLO3, CLO4, CLO5	The student presents slides, gives a demonstration, and answers questions from the lecturer.		End-to-end workflow completion, AI-assisted test case generation, UI/UX finalization	Evaluation Items: - Product (50%): Excellent if all 3 workflows (data setup, main with ≥2 exceptions, dashboard) are integrated and demo runs smoothly; fewer workflows or weak integration lowers the score. - Testing & QA (30%): Excellent with ≥25 test cases (AI + manual), ≥80% passed, full coverage; fewer cases, lower pass rates, or no AI evidence reduce the grade. - Process (20%): Excellent if 3 milestone reports, slides, full10 AI logs, and all members present; missing reports, fewer logs, or weak presentation lead to lower marks. AI Integration: Strong AI usage is required for higher levels across all criteria.	View Rubrics Docume for more details

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
4	Final Project Presentation	Final exam	1	40.0%	4	60'/group	CLO1, CLO2, CLO3, CLO4, CLO5	The student presents slides, gives a demonstration, and answers questions from the lecturer.		Requirement analysing, designing, coding, testing, integrating, AI usage	Final Evaluation The final exam is organized by the Exam Board, with at least two lecturers grading independently. Each team has 60 minutes for presentation and Q&A. Scores from both lecturers will be averaged. Evaluation focuses on: - Requirement Analysis (20%) – Quality of SRS, with AI-assisted requirement elicitation and validation. - System Design (20%) – ERD, state diagrams, architecture, and UI workflows, supported by AI in modeling and refinement. - Software Implementation (40%) – Completeness of workflows, structured codebase, and database design, with AI support in coding and debugging. - AI Usage and Professionalism (20%) – Transparency and responsibility in AI usage (logs, reports, validation of AI outputs), teamwork, and presentation quality.	1. Averag on-going 5/10 to qualify f final 2. Finish main workflow Data Initiation and core function 3. Fail if cheating detected (no resit 4. Respons use of A tools 5. Memb with acti contribu score ≥ 80% — based o peer review, t tracking and cod contribu records.